

# Mnemon: An Agent-Based Study Assistant

with OCR, Retrieval, and Speech

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# Problem Context

**Study is repeated transformation of fragmented course material.**

## STARTING POINT

Slides, scans, and notes

## WHAT CHANGES

Prepare, search, and transform the material

## REVISION FORMATS

Notes, practice, and audio

- Students revise from lecture slides, scans, textbook pages, and personal notes
- The same source must become notes, explanations, flashcards, quizzes, essay practice, or audio
- A plain LLM chat makes fragmented course material harder to manage and answers harder to trace
- The workflow must remain grounded, modular, and inspectable

## Core thesis

Coordinate the workflow around the student's material. Do not treat revision as one unstructured prompt.

# Problem Statement

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## 1. Input readiness

OCR converts scanned documents into text that can be searched and reused.

## 2. Source retrieval

Index course material, then retrieve relevant passages for each student query.

## 3. Output form

Revision requires different outputs: explanation, active recall, essay practice, and listening.

# Aim, Objectives, and Scope

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## Aim and objectives

- Use OCR, retrieval, generation, and speech for common revision tasks
- Keep OCR, RAG, and TTS as separate tools that can be run independently
- Evaluate tool reliability, OCR performance, and recorded workflows

## Scope and limitations

- System-design and engineering thesis
- No claim of measured grade improvement
- No replacement for instructor feedback or student judgement
- Remote model quality, latency, cost, and privacy remain practical constraints

# Study Workflow Requirements

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## OCR

Converts visual material into text.  
OCR errors can affect every later output.

### INSPECTABLE ARTIFACT

OCR cache

## RAG

Retrieves relevant passages from the student's material so answers stay grounded in the source.

### INSPECTABLE ARTIFACTS

RAG ingest file

## TTS

Rewrite diagrams, tables, and formulas for listening, then turn the text into speech.

### INSPECTABLE ARTIFACT

TTS input file

Intermediate artifacts keep each stage visible.

# Architecture: Flexible Agent, Deterministic Tools

The agent chooses the workflow. The tools perform the processing.

## **study-assistant**

Selects one mode: transcription, notes, flashcards, quiz, essay practice, or audio.

## **Tool responsibilities**

Handle caching, chunking, retries, output order, and clear failures.

### **PROCESSING TOOLS**

#### **pdfocr**

Extract text from PDFs

#### **chunkvec**

Store and search material

#### **chunktts**

Create the final audio file

# Engineering Decisions

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## Command-line tools

OCR, retrieval, and speech can be run, inspected, cached, or tested independently.

## Preserve source order

Concurrent requests may complete out of order, but published pages and chunks preserve source order.

## Shared libraries

The tools share common code for HTTP requests and JSON handling.

## Handle partial outputs

OCR keeps successful pages when some requests fail. Incomplete audio is not published.

# Workflow Evaluation Method

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**Evaluate the saved evidence, not only the final response.**

## Evaluation criteria

- Grounding in prepared material
- Fit for the requested study mode
- Saved intermediate outputs
- Reproducible runs
- Visible limitations rather than hidden failures

## Recorded workflows

- **Essay practice:** Association Analysis slides
- **RAG revision:** Naive Bayes textbook section
- **Flashcards:** Anomaly Detection slides
- **Notes to audio:** Clustering notes



# Testing and Recorded Runs

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## What the code controls

- pdfocr: page selection, request IDs, retry queue, JSONL output
- chunkvec: chunk parser, embeddings configuration, SQLite/vector integration
- chunktts: chunk splitting, retry handling, audio validation, final file creation
- relay, jsonx, openai: transport and parsing helpers

## What remote models affect

- Use tests for behaviour controlled by the code
- Use recorded runs for behaviour that depends on remote models
- Check ordering, failures, and tool boundaries
- Record limitations alongside the results

# Evaluation: OCR Throughput

72-PAGE SLIDE PDF BENCHMARK

Running OCR requests concurrently made OCR practical while preserving page order.

**15.89x**

**speedup**

up to 32 concurrent requests

**72/72**

**pages completed**

in both runs

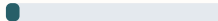
## RUNTIME COMPARISON

One request at a time



**316.66**

Up to 32 concurrent  
requests



**19.93**

Recorded runtime; varies with provider latency, rate limits, network conditions, and document complexity.

# Evaluation: OCR Model Benchmark

## 68 ACADEMIC PAGES FROM 34 PDFS

| Model        | CER   | WER   | Order F1 | Math F1 |
|--------------|-------|-------|----------|---------|
| PaddleOCR-VL | 0.463 | 0.473 | 0.375    | 0.725   |
| olmOCR 2     | 0.468 | 0.489 | 0.325    | 0.674   |
| DeepSeek-OCR | 0.586 | 0.651 | 0.145    | 0.468   |

| Model        | Total cost |
|--------------|------------|
| PaddleOCR-VL | \$0.0420   |
| olmOCR 2     | \$0.0214   |
| DeepSeek-OCR | \$0.0041   |

## Benchmark pages

Hand-labelled academic pages with equations, tables, diagrams, and multi-column layouts.

## Selection tradeoff

olmOCR 2 was less likely to miss important content and cost less than the most accurate model.

# Workflow Evaluation Results

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## Essay practice

Produced strong exam-style questions, but OCR errors make source tracing important.

## Flashcards

Worked well for active recall: concise and examinable, but future checks should cover gaps, duplication, and formula clarity.

## Retrieval

Strongest result: the answer matched the course material, but the retrieved passages should be shown beside it.

## Study notes to audio

Produced audio for revision, but audio cannot fully represent figures, tables, or equations.

# Results and Limitations

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## Interpretation

- Separate tools make the workflow easier to inspect and test
- Concurrent requests reduced OCR runtime in the recorded benchmark
- Retrieval quality depends on how material is split, labelled, and retrieved

## Threats and limitations

- No controlled study of learning outcomes or retention
- TTS evaluation checks the output file, not whether the audio sounds natural
- OCR results depend on the documents and provider
- Model behaviour, pricing, latency, and availability can drift

# Contributions

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## Conceptual

Organises course material into revision formats that remain grounded in the source.

## Technical

Adds ordering, retries, clear failures, and saved outputs at each stage.

## Architectural

Separates the agent from OCR, retrieval, and speech tools.

## Evaluation

Uses tests, benchmarks, and recorded workflows.

# Future Work

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## Next engineering steps

- Record each run from source PDF to final output
- Test retrieval with fixed questions and expected passages
- Test OCR on more documents and repeat the measurements
- Add checks for formulae, diagrams, duplicate cards, and coverage gaps

## Further evaluation

- Compare providers and models for cost, latency, quality, and privacy
- Evaluate local or private inference for sensitive content
- Ask students to evaluate whether the outputs are useful
- Study effects on retention and exam preparation

# Conclusion

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## Main conclusion

study-assistant gives an agent open tools that can be inspected, changed, and run independently.

# 15.89x

**OCR speedup**

recorded OCR benchmark

# RAG

**strongest workflow result**

course-specific answer; retrieved passages should be shown

# TTS

**revision material in audio form**

text is prepared before speech generation



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# Questions

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